Geog4/6300: Lab 4

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One and two sample difference tests

Overview: In this lab, you'll be doing analysis with public use microdata from the Household Pulse survey administered by the U.S. Census Bureau during the COVID pandemic. The survey had a particular focus on pandemic effects on health and wellbeing, but it also included questions on the impact of natural disasters on households. That's the section you'll be focusing on here.

Unlike the aggregate census data you're probably used to working with, microdata provide anonymized individual-level responses that researchers can use to calculate their own summary statistics and estimates.

There's more info on the Pulse survey and the public use microdata file at this link:

- Household Pulse Survey Public Use Files

The main data are contained in the `pulse_2023` csv file in the data folder. The data dictionary for this dataset is also available in the data folder.

You'll use this data to answer two research questions:

- Did households displaced by natural disasters include more children than those not displaced?
- Is there a relationship between household income and a household's ability to return to its place of residence after being displaced?

Assessment criteria

This lab assesses the following learning standards:

21. Assess the normality of a variable and use those results to select between parametric and non-parametric statistical tests.
22. Apply two-sample tests of difference and accurately interpret results in terms of hypothesis testing, statistical significance, and the direction and magnitude of any differences.
23. Construct contingency tables and apply chi-square tests to assess association between categorical variables, interpreting results in terms of significance and practical meaning.

Loading/filtering data and checking for normality.

In the first part of the lab, you'll be using two variables: THHLD_NUMKID (number of children in a household) and ND_DISPLACE (whether a household was displaced by a natural disaster in the last year).

Load the dataset using read_csv, and also take a look at the two variables of interest in the data dictionary. For this lab, you only want observations with valid responses, filtering out records where respondents did not provide an answer. Filter these data so you have only valid responses.

Task 1 *Load the Pulse data and filter it to only valid responses.*

```
pulse_2023<-read_csv("data/pulse_2023.csv")
```

```
Rows: 539305 Columns: 48
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (2): SCRAM, EST_ST
dbl (46): WEEK, EST_MSA, REGION, HWEIGHT, PWEIGHT, TBIRTH_YEAR, ABIRTH_YEAR,...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

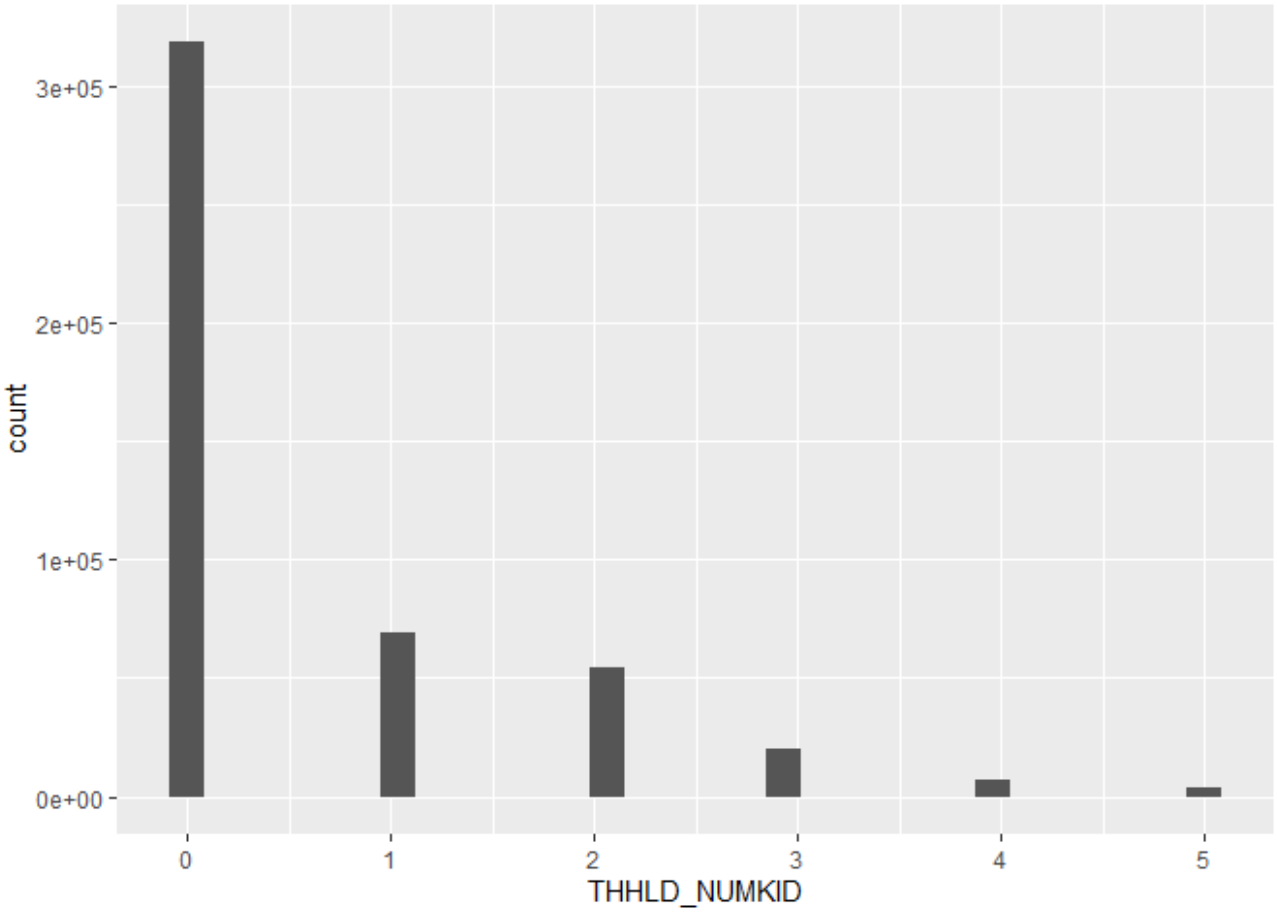
```
pulse_2023_filtered<-pulse_2023 |>
  filter(ND_DISPLACE>=0)
```

You'll be conducting a two-sample test of difference to look at whether the number of children differs between those displaced and not displaced. To know which test to use, you have to assess the normality of the THHLD_NUMKID variable.

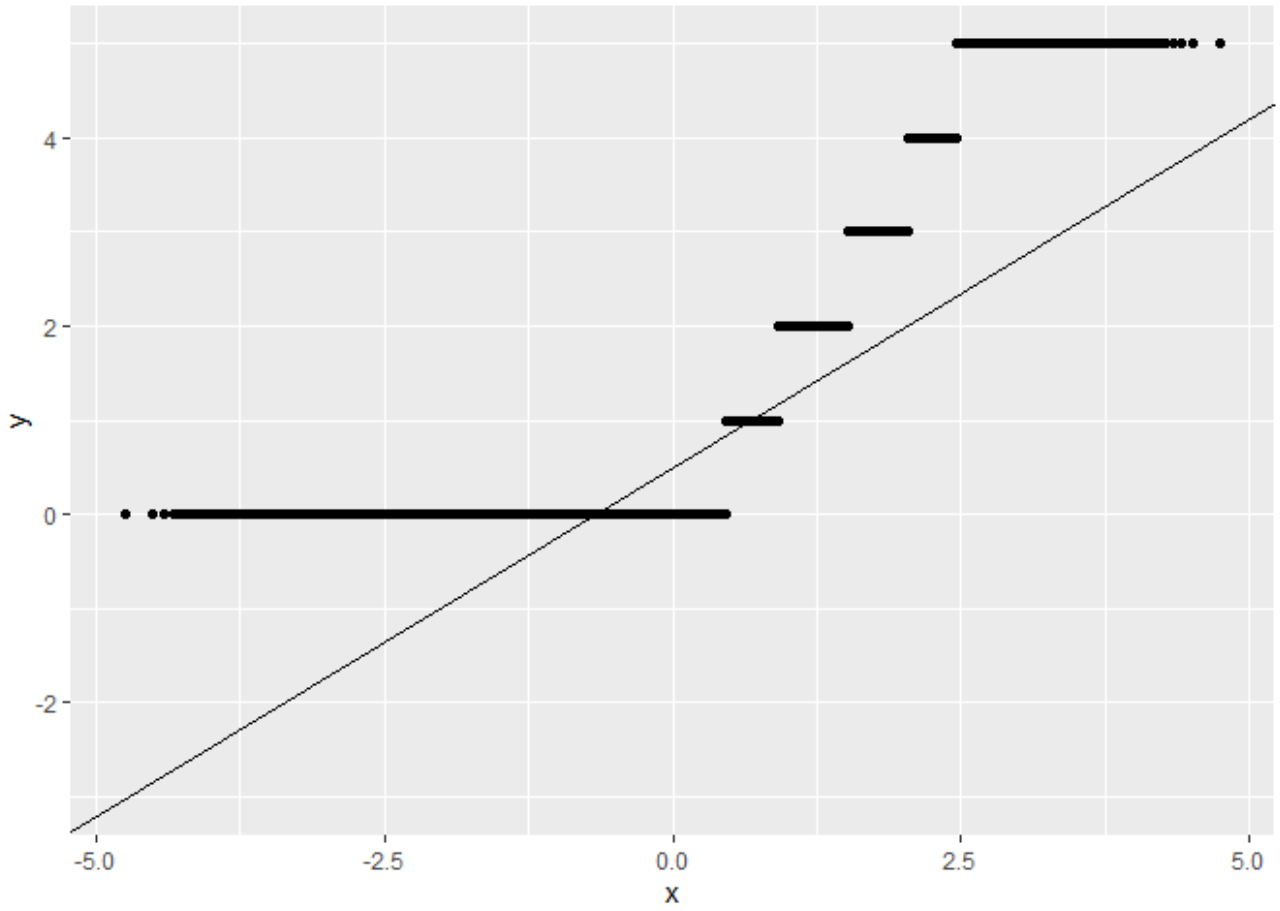
Task 2 *Assess the normality of this variable using either a histogram OR a qq line. Make sure your plot is visible and interpret what it shows you about the normality of this variable.*

```
ggplot(pulse_2023_filtered,aes(x=THHLD_NUMKID)) +
  geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```




```
ggplot(pulse_2023_filtered, aes(sample=THHLD_NUMKID)) +
  stat_qq() +
  stat_qq_line()
```



Testing the difference in number of children.

Now you can test whether the number of children differs between displaced and non-displaced households.

Task 3 *Based on the normality of the data, what is the appropriate statistical test for determining whether the number of children differs between these two groups? Explain why this test is appropriate and give a null and alternative hypothesis for this test*

I would use the Non-parametric (Wilcoxon) t-test. I chose this test because the data is not normally distributed as seen in both the histogram (skews heavily to the right) and the qq plot (not a straight line that follows the qq line).

Null Hypothesis: The number of children between the group that was and group that was not displaced by a natural disaster is not different.

Alternative Hypothesis: The number of children between the group that was and group that was not displaced by a natural disaster is different.

Task 4 *Run the test you identified in Task 3. Also calculate the mean for each dataset, which you can use to determine the nature of the difference–which group (if any) has more children.*

```
wilcox.test(pulse_2023_filtered$THHLD_NUMKID ~ pulse_2023_filtered$ND_DISPLACE)
```

```
    Wilcoxon rank sum test with continuity correction

data:  pulse_2023_filtered$THHLD_NUMKID by pulse_2023_filtered$ND_DISPLACE
W = 1171425061, p-value = 2.463e-12
alternative hypothesis: true location shift is not equal to 0
```

```
pulse_2023_filtered_mean<-pulse_2023_filtered |>
  group_by(ND_DISPLACE) |>
  summarise(mean=mean(THHLD_NUMKID))

kable(pulse_2023_filtered_mean)
```

ND_DISPLACE	mean
1	0.7472826
2	0.5944313

Task 5 *Interpret the results of the analysis you did in task 4.* * Do you reject the null hypothesis? Why or why not? * Based on the test result and the group means, what can we conclude about differences in the number of children between these two groups?

Based on the p-value, which is 2.463e-12 (smaller than 0.05), I reject the null, meaning there is a difference between the two groups. Based on the p-value and the group means, I can conclude that the group that does not get displaced has more children (.15 more from the mean) by a statistically significant margin.

Identifying differences in time displaced by natural disasters across income groups

In Part 2, you used a test of difference that included a continuous and categorical variable. In this section, you'll focus only on households displaced by natural disasters to identify statistically significant differences in time displaced using two categorical variables: one placing responses into groups by income (INCOME) and the other showing the time displaced (ND_HOWLONG).

Again, look at the data dictionary to better understand how these two variables are structured.

Task 6 *Reload the original, unfiltered Pulse dataset (not the one you used in parts 1 and 2). Select only observations for households displaced by a natural disaster using the ND_DISPLACE variable. Then filter for records with valid responses to INCOME and ND_HOWLONG. Call the head of the table with* `kable` *when done.*

```
pulse_2023_Only_Displaced<-read_csv("data/pulse_2023.csv") |>
  filter(ND_DISPLACE=="1" & INCOME>=0 & ND_HOWLONG>=0)
```

```
Rows: 539305 Columns: 48
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr  (2): SCRAM, EST_ST
dbl (46): WEEK, EST_MSA, REGION, HWEIGHT, PWEIGHT, TBIRTH_YEAR, ABIRTH_YEAR,...

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
kable(head(pulse_2023_Only_Displaced))
```

D_UNSANITARY	ND_ISOLATE	ND_CRIME	ND_SCAM	MEDICAID	MEDICAID_
1	1	1	1	3	
1	3	3	4	3	
3	4	1	3	3	
1	2	1	1	3	
1	1	1	4	3	
1	1	1	1	-88	

Task 7 *Create a contingency table of responses to the ND_HOWLONG variable by INCOME. The rows of this table should be the income groups and the columns should be the categorical variables showing time displaced. Call the table when you are done.*

```
Displaced_Table<-pulse_2023_Only_Displaced |>
  count(ND_HOWLONG,INCOME,name="count") |>
  pivot_wider(names_from = ND_HOWLONG, values_from = count)
```

```
kable(Displaced_Table)
```

INCOME	1	2	3	4	5
1	286	218	115	69	179
2	162	122	50	23	63
3	221	135	67	32	50
4	316	160	78	42	52
5	219	107	54	23	33
6	317	115	58	37	37
7	142	54	22	8	11
8	223	64	45	35	43

Task 8 *Run a chi-square test on your contingency table to assess whether there are statistically significant differences in displacement time between these income groups. You'll likely need to delete the INCOME variable to do so.*

```
chisq.test(Displaced_Table |> select(-INCOME))
```

```
    Pearson's Chi-squared test

data:  select(Displaced_Table, -INCOME)
X-squared = 210.51, df = 28, p-value < 2.2e-16
```

Task 9 *To help with your interpretation of results, Calculate the percentage of households within each income group that fall into category 1 of ND_HOWLONG ("displaced less than a week").*

```
Displaced_Table_PCT<-Displaced_Table |>
pivot_longer(`1`:`5`,
             names_to = "ND_HOWLONG",
             values_to = "count") |>
  group_by(INCOME) |> mutate(total=sum(count)) |>
  mutate(pct=round(count/sum(count)*100,2)) |>
  filter(ND_HOWLONG == "1") |>
  select(-count) |>
```

```
  pivot_wider(names_from = ND_HOWLONG, values_from = pct)

#I want to see, separate from the task, what each column would look like
Displaced_Table_PCT_ALL<-Displaced_Table |>
pivot_longer(`1`:`5`,
             names_to = "ND_HOWLONG",
             values_to = "count") |>
  group_by(INCOME) |> mutate(total=sum(count)) |>
  mutate(pct=round(count/sum(count)*100,2)) |>
  select(-count) |>
  pivot_wider(names_from = ND_HOWLONG, values_from = pct)
```

Task 10 *Interpret the results of your analysis in tasks 8 and 9.* * Do you reject the null hypothesis based on these results and why (or why not)? * What do your results in Task 9 tell you about differences in how quickly households were able to return?

I would reject the null hypothesis that there is no difference in how quickly different income categories return home after a natural disaster. I would reject the null hypothesis because the p-value is much lower than 0.05 (2.2e-16). Furthermore, looking at the chi-square distribution table, with a df=28, we would need a X-squared of 41.337 to reject the null. The X-squared value we get from this data is 210.51, so we can reject the null hypothesis using the chi squared table as well.

The results from task 9 tell me that lower income families tend to not return home nearly as quick as wealthy families. Wealthy families return home within a week between 50 and 60 percent of the time, but low income families only do so about 30 percent of the time. I dug further into how long it usually takes low income communities and over 20 percent of the time they don't return home at all. These results overall show that low income families return home in less than a week at nearly have the rate as higher income families.

Challenge Question:

Calculate and map the percentage of households reporting displacement (ND_DISPLACE) by state. To do so, you'll need to start with a version of the dataset that excludes non-valid responses and then calculate the percentage displaced by state.

Once you've calculated that estimate, map it out using the `states_albers` geopackage file in the data folder. Provide a brief description of the patterns you observe.

```
#Code goes here
```

Final submission stuff

Disclosure of assistance: Besides class materials, what other sources of assistance did you use while completing this lab? These can include input from classmates, relevant material identified through web searches (e.g., Stack Overflow), or assistance from ChatGPT or other AI tools. How did these sources support your own learning in completing this lab?

Outside of the class material I had help from Google's AI and 2 classmates. I had to use essentially all 3 to do task 4 because I was confused how we were supposed to test each group from 1 column. Selena helped me by telling me the tilde would group the ND_DISPLACE to either 1 and 2 and sort the THHLD_NUMKID data into them. I then used AI to explore how that was different than the comma version we used in class. Grayson and I also collabed on task 4 briefly, where we discussed the differences in using the comma versus tilde because he was doing what I initially did. After task 4, the only help I needed outside of class material was on task 9, where Grayson and I different looking tables. I missed the chi-squared class where we covered this stuff because I was doing a job shadow at the NWS Greenville-Spartanburg office so while I went back and watched the video, I still needed a bit more help with the percentage table and Grayson helped me understand what went where and why. We collaborated on how to produce the percent table in task 9. Other than those 2 tasks though I just used class notes and scripts for everything else.

Lab reflection: How do you feel about the work you did on this lab? Was it easy, moderate, or hard? What are the biggest things you learned by completing it?

The lab overall was very straightforward. I did not struggle with any task outside of task 4. I found that task a bit frustrating because I don't remember using the ~ before to group data, so I was struggling through that one a fair bit before Selena helped me understand the use of the ~. Other than that 1 task everything else was straightforward. The biggest thing I learned was how efficient it is to run statistical tests in R. If I had known before I was mostly done with my thesis I would have just run the tests in R.